

$m + m - 1 + m - 1 = 3m - 2$
elementi
non
nulli
"
 $O(m)$

Equation i
 $a_{i1}x_1 + \dots + a_{i,i-1}x_{i-1} + a_{ii}x_i + a_{i,i+1}x_{i+1} + \dots + a_{in}x_n = b_i$

JACOBI

$$x_i^{(k+1)} = \frac{b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k)} - \sum_{j=i+1}^n a_{ij}x_j^{(k)}}{a_{ii}} \quad i=1 \dots m$$

COSTO
 $O(m^2 \cdot K)$

convergenza dipende
da Gauss-Jordan se

$m^2 \cdot K \leq \frac{m^3}{3} \rightarrow K \leq \frac{m}{3}$

$a_{i1}x_1 + \dots + a_{in}x_n = b_i$
 $a_{21}x_1 + a_{22}x_2 = b_2$
 $a_{31}x_1 + 0 + a_{33}x_3 = b_3$
 $a_{i1}x_1 + 0 + \dots + a_{ii}x_i = b_i \quad i=2 \dots m$

COSTO

$O((m + 2(m-1)) \cdot K)$

1)

$O(3mK)$

$x_1^{(k+1)} = (b_1 - \sum_{j=2}^m a_{1j}x_j^{(k)})/a_{11}$
 $x_i^{(k+1)} = \frac{b_i - a_{i1}x_1^{(k)}}{a_{ii}} \quad i=2 \dots m$

$A = \begin{pmatrix} \diagup & \diagdown \end{pmatrix}$

CONVERGENZA DI METODI ITERATIVI

CONDIZIONE SUFFICIENTE

A diagonale dominante $\Rightarrow$ Jacobi e Gauss-Seidel convergono
NO DIH

Si dimostra che la matrice associata al problema del page rank è diagonale dominante.

Se A non è diagonale dominante da $\times$ non possiamo dedurre nulla.

METODI ITERATIVI IN FORMA MATRICIALE

JACOBI

$x_i^{(k+1)} = \frac{b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k)} - \sum_{j=i+1}^m a_{ij}x_j^{(k)}}{a_{ii}} \quad i=1 \dots m$

$A = L + D + U = \begin{pmatrix} 0 & 0 \\ a_{ij} & 0 \end{pmatrix} + \begin{pmatrix} a_{11} & 0 \\ 0 & a_{nn} \end{pmatrix} + \begin{pmatrix} 0 & a_{ij} \\ 0 & 0 \end{pmatrix}$

$\begin{pmatrix} 3 & 1 & 2 \\ 4 & 1 & -1 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 3 & 2 & 0 \end{pmatrix} + \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

$x^{(k)} = \begin{pmatrix} x_1^{(k)} \\ \vdots \\ x_n^{(k)} \end{pmatrix} \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \quad x^{(k+1)} = \begin{pmatrix} x_1^{(k+1)} \\ \vdots \\ x_n^{(k+1)} \end{pmatrix}$

$$X = \begin{pmatrix} \vdots \\ x_n^{(k)} \end{pmatrix} \quad b = \begin{pmatrix} \vdots \\ b_m \end{pmatrix} \quad x = \begin{pmatrix} \vdots \\ x_n^{(k+1)} \end{pmatrix}$$

$$X^{(k+1)} = D^{-1} \cdot z$$

$$\begin{pmatrix} 1 & 0 & 0 \\ a_{11} & 1 & 0 \\ 0 & a_{22} & 1 \\ 0 & 0 & 1 \\ & & a_{nn} \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_n \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_n \end{pmatrix}$$

$$x_i^{(k+1)} = \frac{z_i}{a_{ii}} \quad i=1, n$$

$$z \text{ viene t.c.} \quad z_i = b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k)} - \sum_{j=i+1}^n a_{ij} x_j^{(k+1)} \quad i=1, n$$

$$z = b - L \cdot x^{(k)} - U \cdot x^{(k)}$$

$$x^{(k+1)} = D^{-1} (b - L x^{(k)} - U x^{(k)})$$

$$\Rightarrow x^{(k+1)} = \boxed{-D^{-1}(L+U)} x^{(k)} + \boxed{D^{-1} \cdot b}$$

B_J
matrice di iterazione

$$x^{(k+1)} = B_J \cdot x^{(k)} + c$$

LO ABBIAMO
DIMOSTRATO

$$M=D$$

$$N=L+U$$

JACOBI

$$M = \text{diag}(\text{diag}(A))$$

$$N = A - M$$

$$c = M^{-1} \cdot b$$

$$B_J = -M^{-1} \cdot N$$

SI DIMOSTRA CHE GAUSS SEIDEL:

$$x^{(k+1)} = B_{GS} \cdot x^{(k)} + c \quad \text{con } B_{GS} = -M^{-1} \cdot N$$

$$c = M^{-1} \cdot b$$

GAUSS
SEIDEL

$$M = L+D$$

$$N = U$$

$$M = \text{tri}(A)$$

$$N = A - M$$

$$Ax=b$$

$$x^{(k)} \quad k \geq 0$$

$$x^{(0)} \text{ FISSATA}$$

DEFINIZIONE

Un metodo iterativo è convergente $\Leftrightarrow \lim_{k \rightarrow \infty} x^{(k)} = x$

$$Ax=b$$

$$A=M+N$$

$$\text{con } B = -M^{-1} \cdot N \quad c = M^{-1} \cdot b$$

TEOREMA

Un metodo iterativo della forma $x^{(k+1)} = B \cdot x^{(k)} + c$

è convergente $\Leftrightarrow \rho(B) < 1$

$$\rho(B) = \max(|\lambda| : \lambda \text{ eig}(B))$$

DM

metodo convergente $\Leftrightarrow \lim_{k \rightarrow \infty} x^{(k)} = x \Leftrightarrow \lim_{k \rightarrow \infty} e^{(k)} = \vec{0} \Leftrightarrow \lim_{k \rightarrow \infty} B^k \cdot e^{(0)} = \vec{0}$

metodo convergente $\Leftrightarrow \lim_{k \rightarrow \infty} x^{(k)} = x \Leftrightarrow \lim_{k \rightarrow \infty} e^{(k)} = \vec{0} \Leftrightarrow \lim_{k \rightarrow \infty} B^k \cdot e^{(0)} = \vec{0}$

$e^{(k)} = x^{(k)} - x$

$\lim_{k \rightarrow \infty} B^k = 0$ MATRICE NULLA

$$e^{(k)} = x^{(k)} - x = Bx^{(k-1)} + c - (Bx + c) = B(x^{(k-1)} - x) = B \cdot e^{(k-1)}$$

$$\boxed{A = M + N}$$

$$B = -M^{-1}N$$

$$c = M^{-1}b$$

$$Ax = b \Leftrightarrow (M+N)x = b \Leftrightarrow Mx + Nx = b$$

$$\Leftrightarrow Mx = -Nx + b$$

$$\Leftrightarrow x = -M^{-1}Nx + M^{-1}b$$

$$\Leftrightarrow x = Bx + c$$

$$e^{(1)} = B e^{(0)}, e^{(2)} = B e^{(1)} = B \cdot B \cdot e^{(0)} = B^2 \cdot e^{(0)}$$

$$e^{(3)} = B \cdot e^{(2)} = B \cdot B^2 \cdot e^{(0)} = B^3 \cdot e^{(0)}$$

$$\Rightarrow e^{(k)} = B^k e^{(0)}$$

METODO CONVERGENTE $\Leftrightarrow \lim_{k \rightarrow \infty} B^k = 0 \Leftrightarrow \rho(B) < 1$

NO DIM $\rho(B) < 1$

Se B fosse uno scalare:

$$\lim_{k \rightarrow \infty} b^k = 0 \Leftrightarrow |b| < 1$$

generalizzazione
su
matrici

$$B = -M^{-1} \cdot N$$

JACOBI

$$M = \text{diag}(\text{diag}(A))$$

$$N = A - M$$

Gauss-Jordan

$$M = \text{tril}(A)$$

$$N = A - M$$

$$\rho(B) \rightarrow \max(|\text{abs}(\text{eig}(B))|) < 1 \quad \text{converge}$$

$$\geq 1 \quad \text{Non converge}$$